ANSWER KEY — Integration Lesson 13 Test: Real-World Accumulation Problems

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Water flows into a tank at $r(t) = 6t$ liters per minute. How much water enters during the first 3 minutes?

> **Work:** total = $\int_0^3 6t\,dt = [3t^2]_0^3 = 3 \times 9 - 0 = 27$. Units: (L/min) × min = liters. **Answer: 27 liters.**

2. A tutoring business earns at $r(t) = 8 + 2t$ dollars per hour. Total from $t = 0$ to $t = 5$?

> **Work:** $\int_0^5 (8 + 2t)\,dt = [8t + t^2]_0^5 = 40 + 25 = 65$. **Answer: $65.**

3. A colony grows at $r(t) = 3t^2$ bacteria per hour. How many NEW bacteria in the first 4 hours?

> **Work:** new bacteria = $\int_0^4 3t^2\,dt = [t^3]_0^4 = 64 - 0 = 64$. (Antiderivative of $3t^2$ is t^3, since the derivative of t^3 is $3t^2$.) **Answer: 64 new bacteria.**

4. A barrel already holds 15 liters; water flows in at a steady 4 L/min for 6 minutes. Final amount?

> **Work:** added = $\int_0^6 4\,dt = 4 \times 6 = 24$ L. The integral counts only NEW water, so add the start: $15 + 24 = 39$. **Answer: 39 liters.** (Answering 24 = forgot the starting amount.)

5. Sand pours at $r(t)$ kg per minute. What does $\int_1^4 r(t)\,dt$ mean, and its units?

> **Work:** the integral of a rate over a window = total accumulated in that window. Limits 1 and 4 are minutes; units are (kg/min) × min = kg. **Answer: the total kilograms of sand added to the pile between minute 1 and minute 4; units are kilograms.** (Any wording capturing "total sand during that time window" earns the point.)

6. A tank starts empty and fills at $r(t) = 8t$ L/min. When does it reach 100 liters?

> **Work:** amount at time t = $\int_0^t 8u\,du = 4t^2$. Set $4t^2 = 100 \rightarrow t^2 = 25 \rightarrow t = 5$. Check: $4 \times 25 = 100$ ✓. **Answer: at $t = 5$ minutes.**

7. In at 10 L/min, out at $1 + t$ L/min. (a) Net rate. (b) Gain in the first 6 minutes.

> **Work:** (a) net = $10 - (1 + t) = 9 - t$ L/min. (b) gain = $\int_0^6 (9 - t)\,dt = [9t - \tfrac{1}{2}t^2]_0^6 = 54 - 18 = 36$. **Answer: (a) net rate = $9 - t$ L/min; (b) 36 liters.**

8. $r(t) = 2t$ L/min for $0 \le t \le 4$, then constant 8 L/min from $t = 4$ to $t = 7$. Total from 0 to 7?

> **Work:** split at $t = 4$. Phase 1: $\int_0^4 2t\,dt = [t^2]_0^4 = 16$ L. Phase 2: $\int_4^7 8\,dt = 8 \times (7 - 4) = 24$ L. Total $= 16 + 24 = 40$. (Common error: using width 7 for phase 2 → 56, wrong.) **Answer: 40 liters.**

9. Shop A earns a steady \$30/day; Shop B earns 8t \$/day. Over 8 days, who earns more and by how much?

> **Work:** A: $\int_0^8 30\,dt = 240$ dollars. B: $\int_0^8 8t\,dt = [4t^2]_0^8 = 4 \times 64 = 256$ dollars. $B - A = 256 - 240 = 16$. **Answer: Shop B earns more, by \$16.**

10. A dish starts with 100 bacteria, growing at $r(t) = 3t^2$ per hour. When does the population reach 316?

> **Work:** population at time $t = 100 + \int_0^t 3u^2\,du = 100 + t^3$. Set $100 + t^3 = 316 \rightarrow t^3 = 216 \rightarrow t = 6$ (since $6 \times 6 \times 6 = 216$). Check: $100 + 216 = 316$ ✓. **Answer: after 6 hours.**